



Simulating Social Network with LLM Agents: An Analysis of Information Propagation and Echo Chambers

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Abstract. In this paper, we leverage the capabilities of GPT-4o, which is integrated with cognitive modules such as memory, reflection, and chain of thought in simulating social network dynamics. Focusing on the Roe v. Wade case, which significantly influenced abortion laws and women's rights in the United States, we model public opinion propagation on Twitter through likes, retweets, comments, and posts. Multiple simulation experiments are conducted to observe agents over different time intervals. By employing small-world networks and scale-free networks, we analyze the impact of various network structures. Our comparative study reveals variations in interaction patterns and potential echo chamber effects.

Keywords: Social Network Simulation · LLM Agent · Public Opinion Propagation · Echo Chamber Effect · Agent-Based Modeling · Information Dynamics

1 Introduction

The dynamics of social networks have become a significant area of study, particularly with the rise of social media platforms such as Twitter, which significantly influence public opinion and behavior [1, 2]. Recent advancements in natural language processing (NLP) and large language models (LLMs) like GPT-4 offer new possibilities for simulating and analyzing these complex interactions [3, 4]. By integrating cognitive modules such as memory, reflection, and chain of thought, these models can more accurately mimic human decision-making processes [5]. This paper explores the use of these advanced models to simulate public opinion propagation, focusing on the Roe v. Wade debate as a case study.

In this paper, we model the mechanisms of public opinion propagation on Twitter, including likes, retweets, comments, and posts, using the Roe v. Wade case as an example. By conducting multiple simulation experiments, we observe agents' actions, attitudes, and attitude scores over various time intervals. To

understand the impact of different network structures, we employ small-world networks and scale-free networks as control variables [6–8]. Our approach allows for a controlled examination of how information disseminates through these networks, providing insights into interaction patterns and potential echo chamber effects [9].

Our study includes experiments to explore the dynamics of information spread and echo chamber formation in detail. By employing 15 agents over 10 timesteps, with each timestep representing half a day, we compare results to identify key differences in interaction patterns and how network structures affect information dissemination.

Understanding how public opinion spreads through different networks is significant for policymakers and social media platforms, as it can inform strategies to manage information and reduce misinformation [2, 10]. Integrating advanced NLP models into social network simulations offers new interdisciplinary research opportunities that combine computer science, psychology, and sociology [11]. Hence, this paper not only advances theoretical understanding of social networks but also provides practical applications to improve information management in social networks.

This paper is structured as follows: Sect. 2 reviews the relevant literature on social network analysis. Section 3 details the methodology, including the simulation frameworks and the incorporation of cognitive modules in LLM agents. Section 4 describes the experimental setup, network initialization, agent interactions, and simulation parameters. Section 5 presents the results of our simulations, including visualizations of opinion dynamics, echo chamber effects, and attitude changes. Section 6 discusses the findings and their implications for information dissemination in social networks.

2 Related Work

Recent advancements in LLMs have significantly expanded the capabilities of agent-based modeling (ABM). The integration of LLMs into ABM frameworks has enabled the simulation of complex agent behaviors and interactions within various domains, enhancing the realism and applicability of these models.

Brown et al. (2020) further showcased the potential of LLMs with their work on GPT-3, highlighting the models’ capabilities in few-shot learning scenarios [3]. This advancement has created new opportunities for developing agents capable of understanding and producing human-like text, making them more effective in simulating realistic interactions. Park et al. (2023) in their innovative study present a groundbreaking application of LLMs in creating a simulated environment known as “Stanford Town” [12]. This environment features generative agents that exhibit human-like behaviors and interactions, grounded in contextually rich scenarios.

Building on these foundational advancements, modern research has increasingly used LLMs to simulate complex agent behaviors and interactions in diverse fields. Notably, Yurkhymentko et al. (2022), in their work present a synthetic

dataset, SynthPAI, designed to address privacy risks in personal attribute inference by employing LLMs to generate synthetic profiles and interactions on Reddit [13]. Similarly, Li et al. (2023) evaluate the proficiency of LLMs in generating counterfactual scenarios across various natural language understanding tasks [14].

The potential of LLMs for large-scale social simulations is shown in Mou et al. (2023) with their study [15]. This work uses LLM-powered agents to simulate social movement dynamics, giving insights into how social change and mobilization happen on digital platforms. Further advancing this field, Vezhnevets et al. (2023) uses LLMs to create actions that fit various environments [16]. Concordia demonstrates how using LLMs in agent-based modeling (ABM) can make simulations richer, providing a flexible tool for studying complex systems and agent behavior.

Furthermore, Gao et al. (2023) provide an extensive review of the current state and future directions of integrating LLMs into ABM [17]. The challenges and opportunities in areas such as environment perception, human alignment, action generation, and evaluation are highlighted by the survey. The latest works in various domains—including network, physical, social, and hybrid environments—are also reviewed, emphasizing the potential of LLMs to enhance the realism and applicability of ABM in both real-world and virtual settings.

Compared to previous related work, we mainly focus on the application value of LLM in social network simulation. By creating relevant agents and various social network structures, we apply LLM simulations to topics in social sciences such as information dissemination and the echo chamber effect. We provide several well-designed control experiments and visualizations of simulations, offering new insights for social network simulation.

3 Methodology

3.1 Overview

In this study, we utilize advanced NLP models and simulation frameworks to investigate the propagation of public opinion on social networks. Our approach builds on the HiSim¹ and AgentVerse² frameworks [18], which provide robust platforms for simulating multi-agent interactions and social network. This integration allows us to simulate agents with sophisticated cognitive abilities, including memory, reflection, and chain of thought, enhancing the realism and depth of our simulations.

3.2 Large Language Model Agents

Based on the work by Cheng et al. (2024), we incorporate LLM agents into our simulations [19]. These agents are designed to exhibit advanced cognitive

¹ <https://github.com/xymou/HiSim>.

² <https://github.com/OpenBMB/AgentVerse>.

functions, which enhance their ability to simulate human-like interactions in social networks. The key cognitive modules integrated into these agents include:

Memory: This module enables agents to retain and recall past interactions and information, which influences their future decisions and behaviors. By simulating memory, agents can develop more consistent and context-aware responses, closely mimicking human behavior.

Reflection: Reflection allows agents to evaluate their actions and adjust their strategies based on outcomes and feedback from the environment. This capability is crucial for modeling adaptive behaviors and understanding how individuals learn and evolve over time.

Chain of Thought: This module facilitates complex reasoning processes, enabling agents to make more informed decisions by considering multiple factors and potential consequences. It allows for more sophisticated decision-making, reflecting the nuanced thought processes humans use in real-world scenarios.

These cognitive capabilities, powered by LLMs like GPT-4o, significantly enhance the realism and depth of our simulations, providing deeper insights into the dynamics of public opinion propagation on social networks.

3.3 Simulation Frameworks

HiSim. HiSim is a social media simulation framework that categorizes users into two types: core users driven by LLMs and numerous ordinary users modeled by deductive agent-based models. HiSim allows for the customization of agent behaviors and network structures, providing a versatile platform for studying the dynamics of information dissemination. In our study, HiSim is used to create various network configurations, including independent agents, small-world networks, and scale-free networks, to examine how these structures influence public opinion propagation.

AgentVerse. AgentVerse is designed to facilitate the deployment of multiple LLM-based agents in various applications, primarily providing two frameworks: task-solving and simulation. It extends the capabilities of HiSim by integrating advanced NLP models, enabling the simulation of agents with more complex cognitive functions. This framework supports the implementation of LLMs like GPT-4, allowing agents to engage in more realistic and context-aware interactions. By leveraging AgentVerse, we can model agents' decision-making processes more accurately, reflecting real-world scenarios where individuals' opinions and actions are influenced by their cognitive abilities and social contexts.

4 Experimental Design and Procedure

The flowchart (Fig. 1) provides a visual representation of each step involved in the experiment, ensuring a clear understanding of the process.

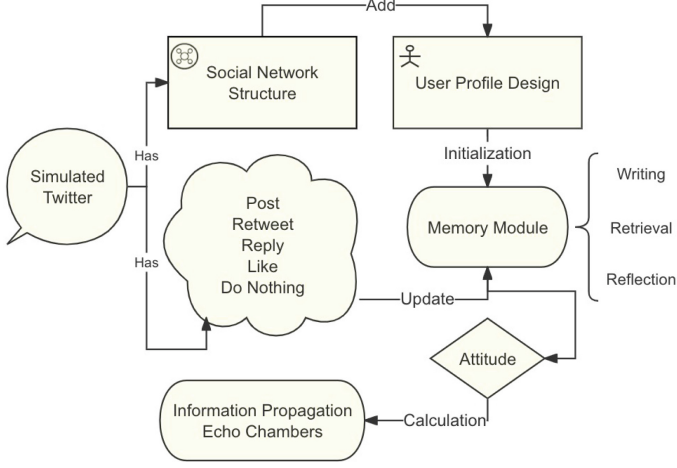


Fig. 1. Framework of the Experimental Design and Procedure. First, we set up a simulated Twitter environment. Second, the social network structure and user profiles are designed and initialized. Third, users interact through various actions such as posting, retweeting, replying, liking, or doing nothing. Finally, the memory module updates user attitudes, and information propagation and echo chambers are analyzed.

4.1 Experimental Setup

To explore the effects of different network structures and agent configurations on public opinion dynamics, we conduct two sets of experiments. Each setup is designed to simulate real-world social network interactions and observe the resulting opinion propagation and echo chamber formation.

Network Initialization:

Scale-Free Network: We initialize a network where a few nodes (agents) have a high number of connections, while most nodes have relatively few. This setup simulates real-world social networks with influential hubs.

Small-World Network: We create a network with high clustering coefficients and short average path lengths, representing tightly-knit communities with occasional long-range connections.

Agent Initialization:

Cognitive Capabilities: Each agent is equipped with advanced cognitive modules, including memory, reflection, and chain of thought, to simulate human-like interactions.

Initial Opinions: The initial opinions of agents are determined by the first post generated by GPT-4 based on the background settings of each agent. These posts are crafted to reflect a diverse range of viewpoints and serve as the starting point for opinion propagation.

Simulation Parameters:

Timesteps: The simulations are run over a series of timesteps to observe the evolution of opinions and the formation of echo chambers. Our setup involves 15 agents over 10 timesteps, with each timestep representing half a day, allowing for a continuous and expanded analysis of opinion dynamics.

Interaction Rules: Agents interact with their neighbors according to rules designed to mimic Twitter-like behaviors. Each round, agents can choose to retweet, post, like, or comment. These behaviors and their outcomes are determined using OpenAI’s function call rules and detailed prompts.

4.2 Experimental Procedure

Initialization:

The network structure is initialized by gpt-4o, and agents are assigned their initial opinions and cognitive capabilities.

We referred to Mou et al.’s study to design prompts for simulating interactions on Twitter [15].

Prompts:

Now you are acting as an agent named `#{agent_name}` in the social media Twitter. You might need to performing reaction to the observation. You need to answer what you will do to the observations based on the following information:

- (1) The agent’s description: `#{role_description}`
- (2) Current time is `#{current_time}`
- (3) The news you got is ‘`#{trigger_news}`’
- (4) Your history memory is `#{personal_history}`
- (5) Your recent memory is `#{chat_history}`
- (6) The twitter page you can see is `#{tweet_page}`
- (7) The notifications you can see are `#{info_box}`

Simulation Execution:

Interaction Phase: At each timestep, agents interact with their neighbors. During these interactions, they can choose to retweet, post, like, or comment based on the information they receive and their cognitive processes.

Retweet: Agents share a post from another agent, spreading the information further within the network.

Prompts:

Retweet or quote an existing tweet in your twitter page.
‘content’ is the statements that you add when retweeting.
If you want to say nothing, set ‘content’ to None.
‘author’ is the author of the tweet that you want to retweet,
it should be the concrete name.
‘original_tweet_id’ and ‘original_tweet’ are the id and
content of the retweeted tweet.

Post: Agents create a new post based on their current opinion and the information they have accumulated.

Like: Agents express approval of a post, which may influence the opinions of others who see the liked post.

Comment: Agents add their thoughts to a post, potentially influencing the original poster and other commenters.

Opinion Update: After each interaction, agents update their opinions based on the received information, their cognitive capabilities, and their memory of past interactions. This includes an automatic evaluation of attitude inclination and a 0–1 attitude score generated by a LLM.

Data Collection: Throughout the simulation, data on agents’ interactions, opinion changes, and attitude scores are collected for analysis.

Data Analysis:

Opinion Spread: The extent and speed of opinion dissemination across the network are measured. This involves tracking how quickly opinions spread from one agent to another and the overall reach of specific opinions.

Echo Chamber Formation: The degree of clustering of similar opinions and the isolation of differing opinions are analyzed. This includes identifying close groups of like-minded agents and measuring the extent of opinion polarization.

Attitude Analysis: The overall attitude of the network and its evolution over time are evaluated. Attitude scores are calculated for each agent, and trends in attitude changes are identified.

5 Results of Simulations

5.1 Overview

This section presents the results from our simulation experiments conducted on Twitter regarding the *Roe v. Wade* case, which was an important Supreme Court decision in 1973 that allowed abortion in the United States. The case was initiated by “Jane Roe” (a pseudonym for Norma McCorvey), who challenged Texas laws restricting abortion. The Court decided that those laws went against the Constitution, accepting a woman’s right to privacy, which includes the choice to have an abortion. This decision has a big impact on women’s rights and has been a key point in legal and political talks.

The simulations are performed using agents built with GPT-4o, incorporating advanced cognitive modules such as memory, reflection, and chain of thought. We conduct a comparative analysis utilizing two distinct network structures: small-world networks and scale-free networks. The primary metrics observed are interaction patterns, attitude scores, and the presence of echo chamber effects across different time intervals.

The visualizations in Figs. 2 and 3 show how attitude scores change over time in the network graphs. Each subplot represents a single time step in the simulation. The size and color intensity of the nodes reflect the attitude scores of individual agents; larger and green nodes indicate higher attitude values.

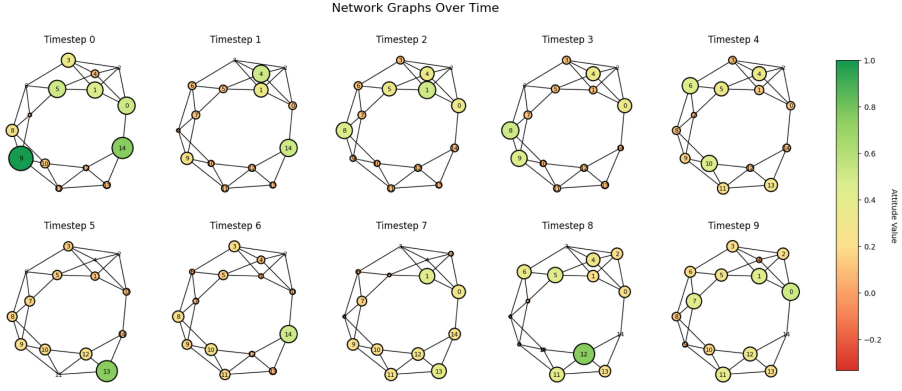


Fig. 2. Diagram of Attitude Evolution in a Small-World Network Structure with 15 Agents. (Color figure online)

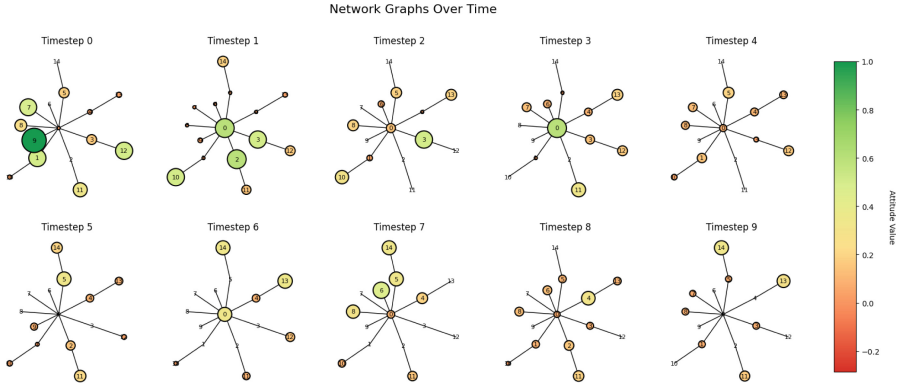


Fig. 3. Diagram of Attitude Evolution in a Scale-Free Network Structure with 15 Agents. (Color figure online)

Small-world social networks in the experimental group exhibit a more diverse and dynamic interchange of opinions and attitudes compared to other structures. This is attributed to lower clustering coefficients and a reduced potential for echo chamber effects, which contribute to higher opinion diversity and elevated average attitude scores. In contrast, scale-free networks highlight the significant influence of central nodes on overall attitude trends. Once these central nodes cease to update, the overall network attitude tends towards neutrality. These visualizations provide deeper insights into how attitude changes in different network structures, helping us better understand online social behaviors.

By comparing the small-world networks and scale-free networks to the control group of independent agents, we observe that isolated agents, except those with strong topic relevance, show minimal changes in their attitude scores, suggesting that individual agent biases are negligible in the networked context. The

small-world network facilitates more frequent and diverse opinion exchanges, highlighting its unique interaction dynamics.

5.2 Echo Chamber Effects

The analysis of echo chamber effects, essential for understanding the dynamics of opinion propagation and attitude polarization, was rigorously examined using metrics on clustering coefficients and a specific echo chamber effect function from the literature.

Clustering Coefficient:

In our simulations, we find that the clustering coefficient is much lower in small-world networks compared to other network structures. This means there are fewer tightly-knit groups of nodes, promoting broader interactions instead of isolated conversations. The random long-range connections in small-world networks likely help create these diverse interactions (see Fig. 4).

Echo Chamber Effect Function:

Huang and Tang [20] propose an echo chamber effect function to evaluate the degree of polarization and the level of homogeneous interactions. This function considers whether users prefer to interact with others who share similar or identical opinions. They define it using three key indicators: the polarization coefficient, the stance expectation coefficient, and the homogeneous interaction coefficient. We adopt their approach to analyze the echo chamber effect in social networks.

The polarization coefficient describes the distribution of different stances within a group, denoted as $F(\cdot)$. For instance, if the distribution of stances on a topic is significantly polarized between positive and negative, $F(\cdot)$ will exhibit a bimodal distribution. The distribution of a stance s can be calculated by its proportion $N^{T,s}$ within the total N_T , T represents the topic, and s represents the stance. N_T is the set of users related to topic T , $N^{T,s}$ is the set of users with stance s regarding topic T .

$$F(T, s) = \frac{|N^{T,s}|}{|N^T|} \quad (1)$$

The stance expectation coefficient quantifies the degree of polarization within a group sharing similar views on a topic T , effectively capturing the expected distribution of stances. Here, $o(T, s)$ represents the inclination value of users towards topic T in the group with stance s , while o_i^T denotes the stance inclination of user i on topic T .

$$p(T, s) = E[o(T, s)] = \frac{\sum_{i \in N^T} o_i^T}{|N^T|} \quad (2)$$

The homogeneous interaction coefficient measures the degree of interaction among users with similar stances by assessing the distance between a user's

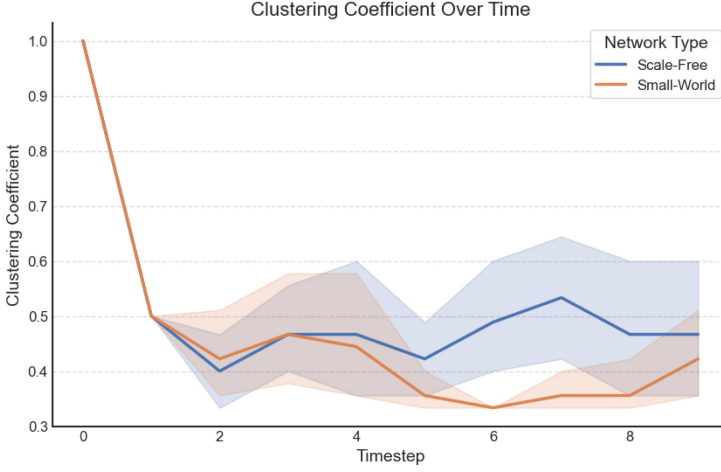


Fig. 4. The changes in the average clustering coefficient over timesteps for two network types, with the shaded area representing the confidence interval.

stance inclination o_i^T and that of their interacting neighbors u_i^T . In echo chambers, users tend to interact more with others holding similar views, and the proximity between o_i^T and u_i^T can be quantified using p -norm $\|o_i^T - u_i^T\|_p$. Hence, the overall homogeneous interaction level for topic T within the group can be represented by the average stance distance between users and their neighbors.

$$i(T) = E[\text{distance}(o, u)] = \frac{\sum_{i \in N^T} \|o_i^T - u_i^T\|_p}{|N^T|} \quad (3)$$

denotes the average stance distance for topic T .

Since the primary stance states in user interaction networks are -1, 0, and 1, calculating expectations across all stance states simultaneously might yield incorrect results. For example, a network perfectly polarized into stances 1 and -1 might still exhibit an overall stance expectation of 0. Therefore, we employ a piecewise function to categorize stances s into three types based on user $o_i \in [-1, 1]$ and utilize an exponential function to smoothly integrate the three aforementioned indicators.

$$f(T) = \begin{cases} f(T, +1) \\ f(T, 0) \\ f(T, -1) \end{cases} = \begin{cases} F(+1) \cdot p(+1) \cdot \exp\{-i(+1)\}, o_i \in (0.5, 1] \\ F(0) \cdot (1 - p(0)) \cdot \exp\{-i(0)\}, o_i \in (-0.5, 0.5] \\ F(-1) \cdot p(-1) \cdot \exp\{-i(-1)\}, o_i \in [-1, -0.5] \end{cases} \quad (4)$$

Our results show that small-world networks yielded a substantially lower score on this function. This suggests that small-world networks are less prone to trapping opinions within homogeneous subgroups, thereby allowing a more varied and dynamic interchange of perspectives (see Fig. 5).

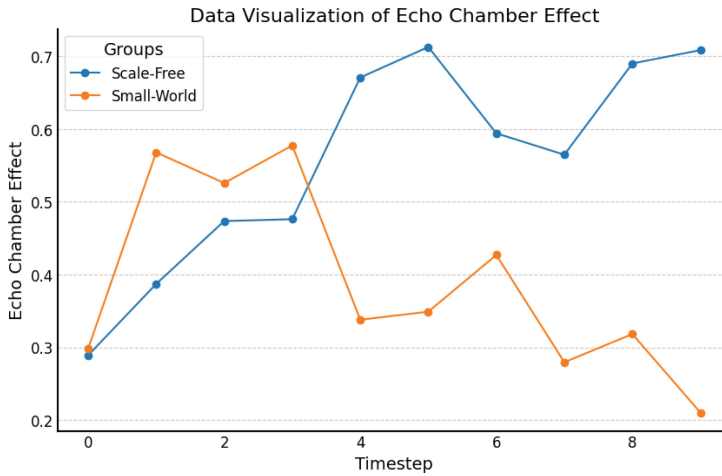


Fig. 5. The changes in the echo chamber effect over timesteps for two network types.

These findings underscore the reduced potential for echo chambers in small-world networks, promoting greater opinion diversity and interaction frequency compared to other network structures. The comparatively lower clustering coefficient and echo chamber effect scores highlight the unique advantage of small-world networks in mitigating homogeneity and fostering a richer discourse environment.

5.3 Changes in Average Attitude and Opinion Diversity

This subsection explores the evolution of average attitude and opinion diversity within the different network structures over time.

Average Attitude:

Our simulations reveal distinct patterns in average attitude across the network types. In small-world networks, average attitude scores exhibit significant fluctuations over time, but they do not trend towards neutrality. This pattern suggests ongoing dynamic interactions and a continuous exchange of diverse opinions, preventing attitude from stabilizing at a neutral point. In contrast, the scale-free networks exhibit a stabilization of attitude scores towards neutrality, especially when the central nodes cease updating their opinions. This indicates a strong dependency on the activity of these key nodes for maintaining dynamic attitude (see Fig. 6).

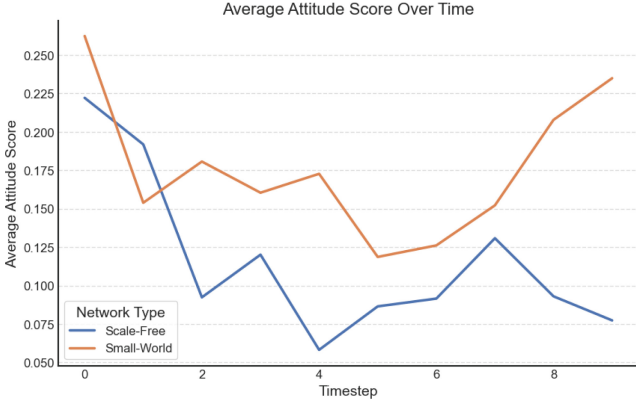


Fig. 6. The changes in the average attitude scores over timesteps for two network types.

Opinion Diversity. The diversity of opinions is another critical metric analyzed in our study. In small-world networks, opinion diversity remains relatively high throughout the simulation period. This sustained diversity is likely a result of the decentralized interaction patterns that prevent the formation of echo chambers. Conversely, scale-free networks, despite their robust connectivity, show a decline in opinion diversity over time, pointing to a potential convergence towards dominant opinions influenced by central nodes (see Fig. 7).

The comparative analysis of average attitude and opinion diversity illustrates the unique dynamics within each network structure. Small-world networks' ability to maintain dynamic and varied interactions contributes to perpetually fluctuating attitudes and sustained opinion diversity. Meanwhile, the scale-free networks' dependence on central nodes underscores their vulnerability to opin-

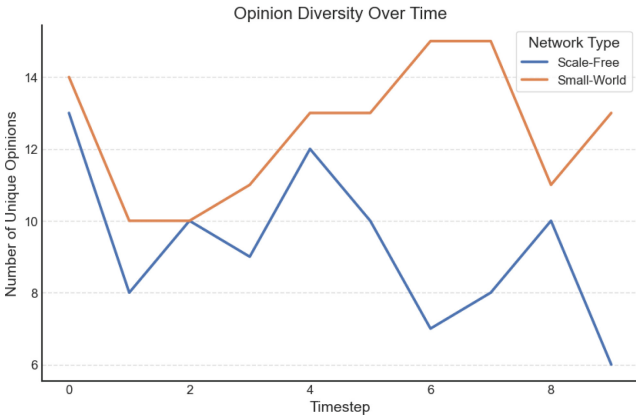


Fig. 7. The changes in opinion diversity over timesteps for two network types.

ion homogenization when activity is concentrated among few influential agents. These insights emphasize the importance of network structure in shaping the landscape of social interactions and opinion formation.

6 Conclusion

In this paper, we employ GPT-4o integrated with advanced cognitive modules to simulate public opinion dynamics on Twitter, specifically in the context of the *Roe v. Wade* case. By creating simulations across different network structures—small-world networks and scale-free networks—we aim to uncover how these structures influence attitude propagation, opinion diversity, and the presence of echo chamber effects.

Our key findings are as follows:

Clustering Coefficient and Echo Chamber Effects: Small-world networks demonstrate significantly lower clustering coefficients and echo chamber effect scores compared to other network structures. This characteristic allows for a more diverse and widespread flow of information, reducing the likelihood of homogeneous groupthink and fostering a richer exchange of opinions.

Attitude Dynamics: Attitude analysis over time reveals that small-world networks maintain fluctuating attitude scores without stabilizing at neutrality, indicating ongoing dynamic interactions. In scale-free networks, attitude scores tend to neutralize when central nodes cease to update, highlighting the critical role of influential nodes in these networks.

Opinion Diversity: Small-world networks consistently exhibit higher opinion diversity throughout the simulation period. This indicates that their unique structure can sustain varied viewpoints and prevent convergence towards a single dominant opinion. Conversely, scale-free networks experience a decline in diversity over time, suggesting a susceptibility to opinion homogenization driven by central nodes.

Control Group Comparisons: Independent agents, used as a control group, display minimal changes in attitude unless the agents have a strong pre-set relevance to the topic. This reinforces the minimal impact of intrinsic agent biases in the absence of network influence, underscoring the importance of network structure in shaping opinion dynamics.

The implications of these findings are substantial for understanding information dissemination in social networks. Small-world networks, with their lower clustering and reduced echo chamber effects, demonstrate potential advantages in fostering a more dynamic and diverse discourse environment. Scale-free networks, while robust in connectivity, are more susceptible to opinion homogenization, emphasizing the critical influence of key nodes.

These findings show the importance of controlled simulations in social network research and help us understand how information spreads and attitudes change in complex social systems. Based on the findings of this study, future research may explore more variables and real-world applications.

In conclusion, our research bridges the gap between advanced NLP technologies and social network analysis, providing a novel framework for studying public opinion dynamics. The capabilities of models can be leveraged to achieve greater accuracy and depth in simulating and analyzing complex social interactions. Future work will focus on refining these models and extending the simulations to larger and more diverse networks, potentially incorporating real-time data to enhance the realism and applicability of the findings [21]. This approach not only enriches our understanding of social networks but also offers practical insights for managing information flow in an increasingly connected world.

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